



WELCOME

Hospital Management System.

Members:-

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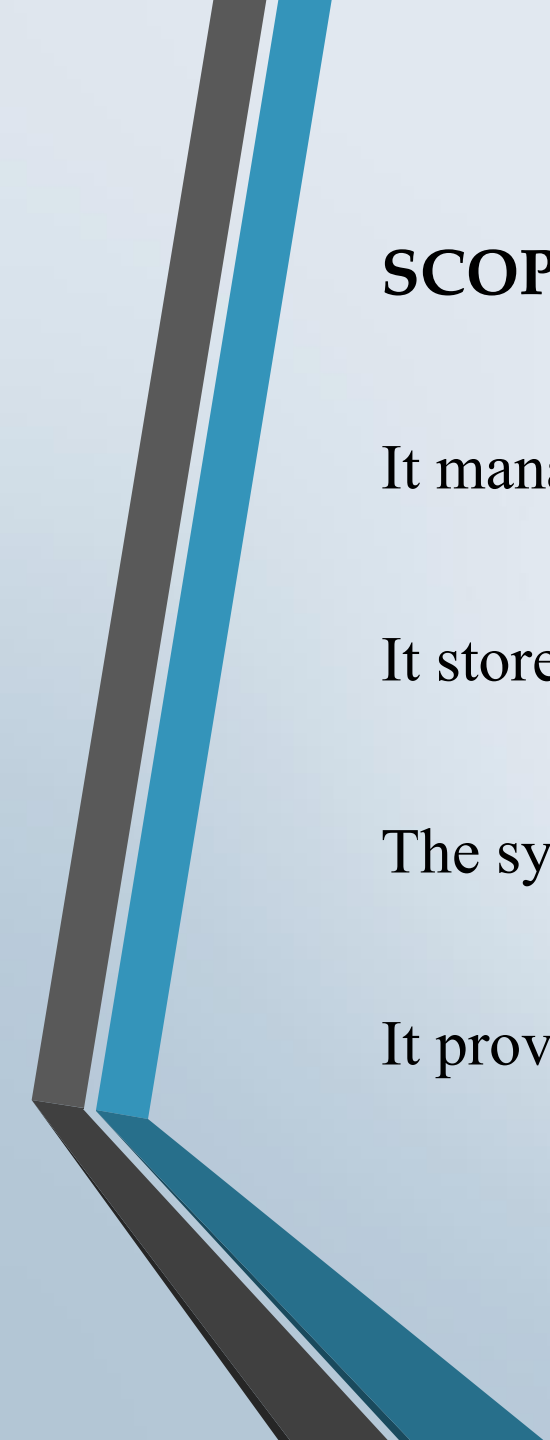
Project name:- Hospital Management System for Shingate Hospital.

Address:- Shingate Hospital Indapur.

Introduction

The **Hospital Management System** is a Java-based desktop application designed to manage hospital operations effectively. It helps in maintaining patient details, doctor information, room availability, billing records, and discharge summaries in a digital format. The system aims to replace manual record-keeping with an automated solution that is faster, more accurate, and reliable.

This system ensures that hospital staff can quickly register patients, assign doctors, generate bills, and retrieve information easily. It improves coordination among different departments and enhances the overall efficiency of hospital administration.



SCOPE:

It manages **patient registration, doctor details, room allotment, and billing.**

It stores all records in a **MySQL database** for easy access and updating.

The system can be used in **small to medium-sized hospitals or clinics.**

It provides a **user-friendly interface** using **Java Swing** for staff to operate easily.

Objectives:-

- To automate hospital operations and reduce paperwork.
- To maintain accurate and organized patient and doctor records.
- To generate bills and reports quickly and efficiently.
- To ensure secure storage of hospital data using a database.
- To improve the coordination between hospital departments and reduce manual errors.

Existing system:-

In the existing (manual) system, most hospital operations are performed on paper. Patient records, doctor schedules, and billing are handled manually, which leads to:

- Time-consuming processes.
- Difficulty in finding old records.
- Chances of human error.
- Poor coordination between departments.
- Data loss due to misplaced files or damage.

Proposed system:-

- **Patient Management** – Register new patients and update their details.
- **Doctor Management** – Store doctor information and assign them to patients.
- **Room Management** – Allocate rooms based on availability and type (General or ICU).
- **Billing Module** – Automatically calculate and generate patient bills.
- **Search and Update** – Quickly search or edit patient and doctor data.
- This system increases hospital productivity and ensures better patient service through fast access to accurate information.

Hardware Requirements :-

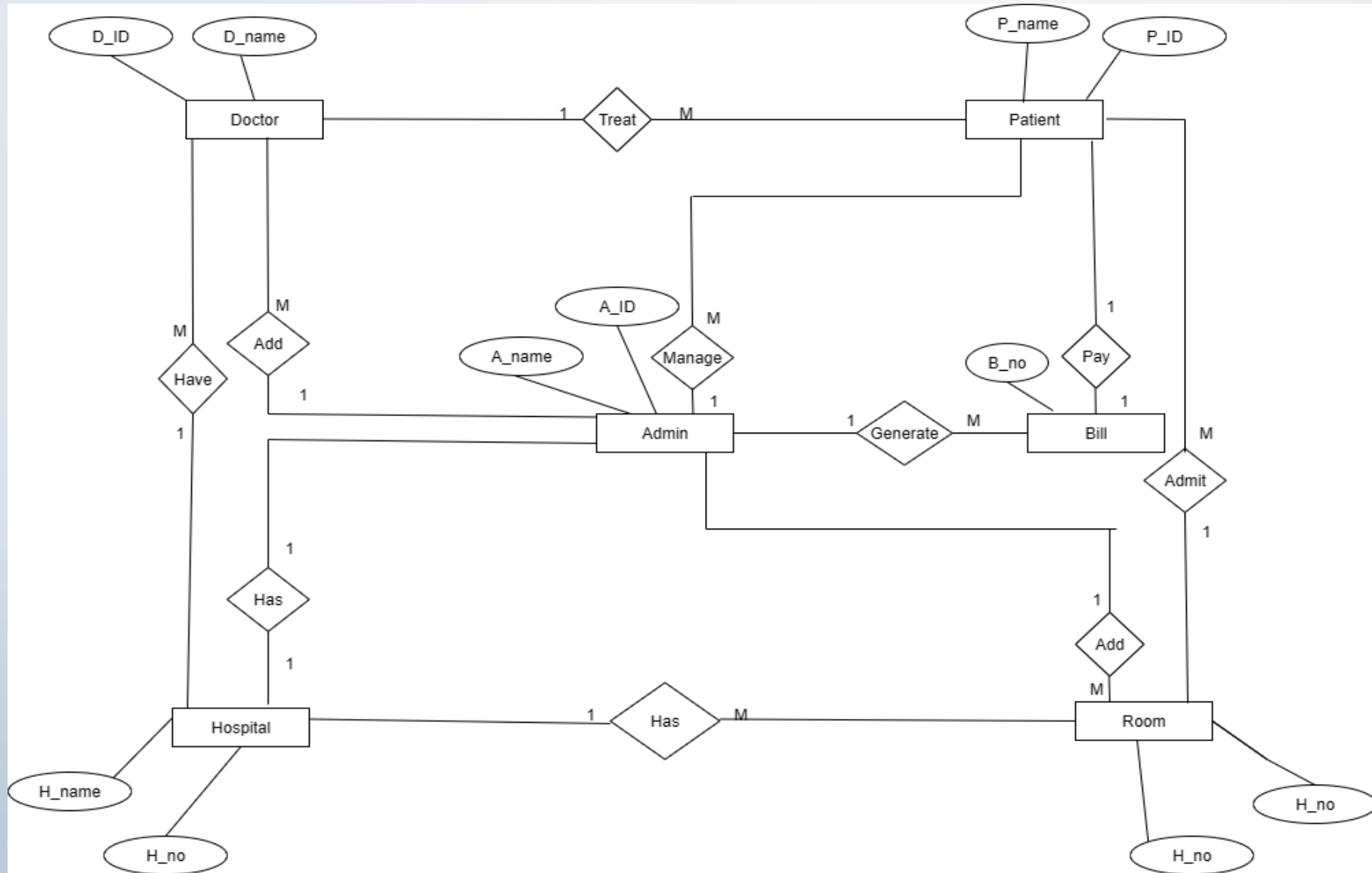
- 1)Processor : Standard processor with a speed of 1.6 GHz or more
- 2)RAM : 2 GB RAM.
- 3)Hard Disk : 500 GB .
- 4)Interface: keyboard ,Mouse.

Software Requirements :-

- 1)Operating System : Windows.
- 2)Front End :Java (Swing),
- 3)Program Language : Java
- 4)Back End : MYSQL.
- 5)Software: XAMP, IDE: NetBeans

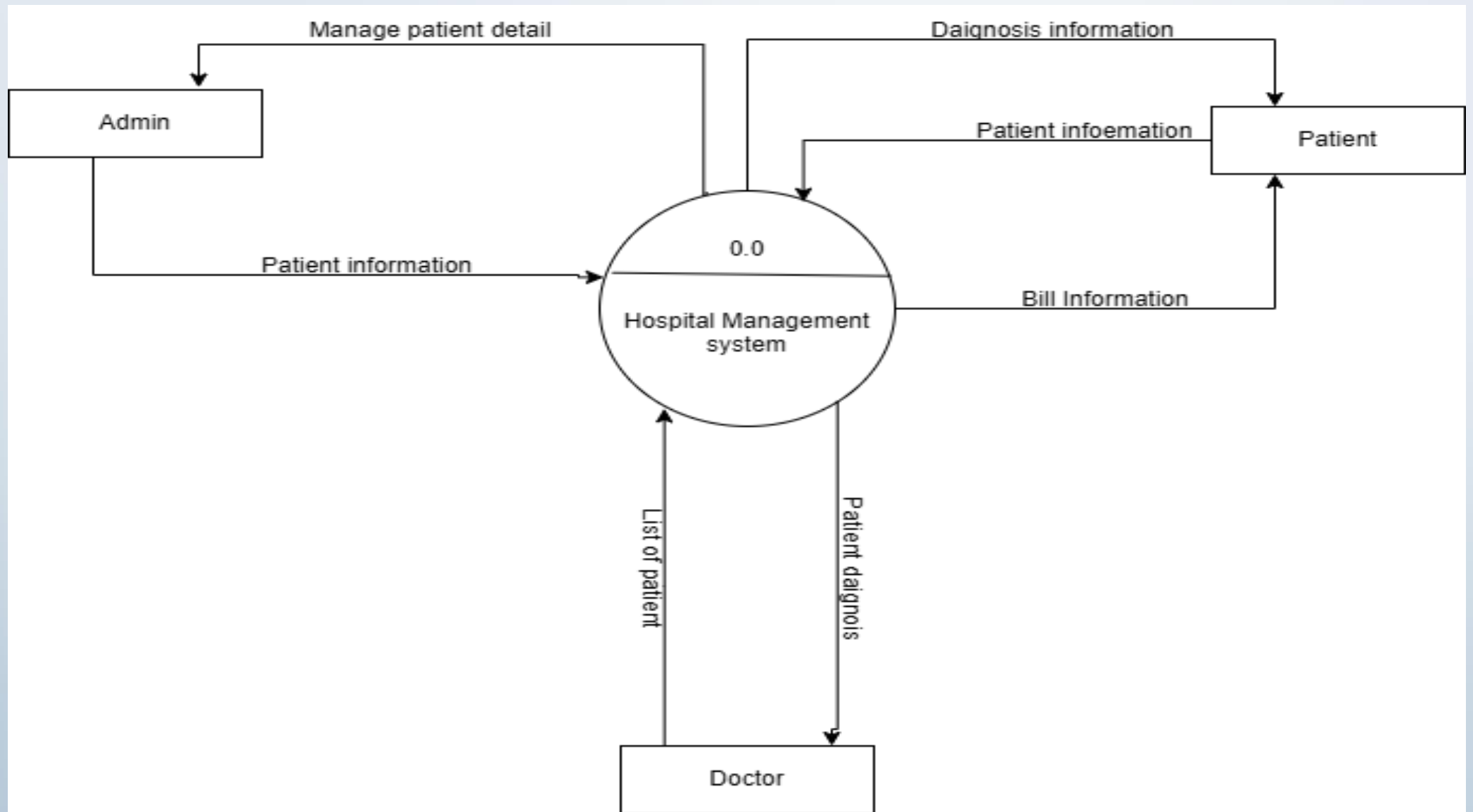


Entity relation ship Diagram



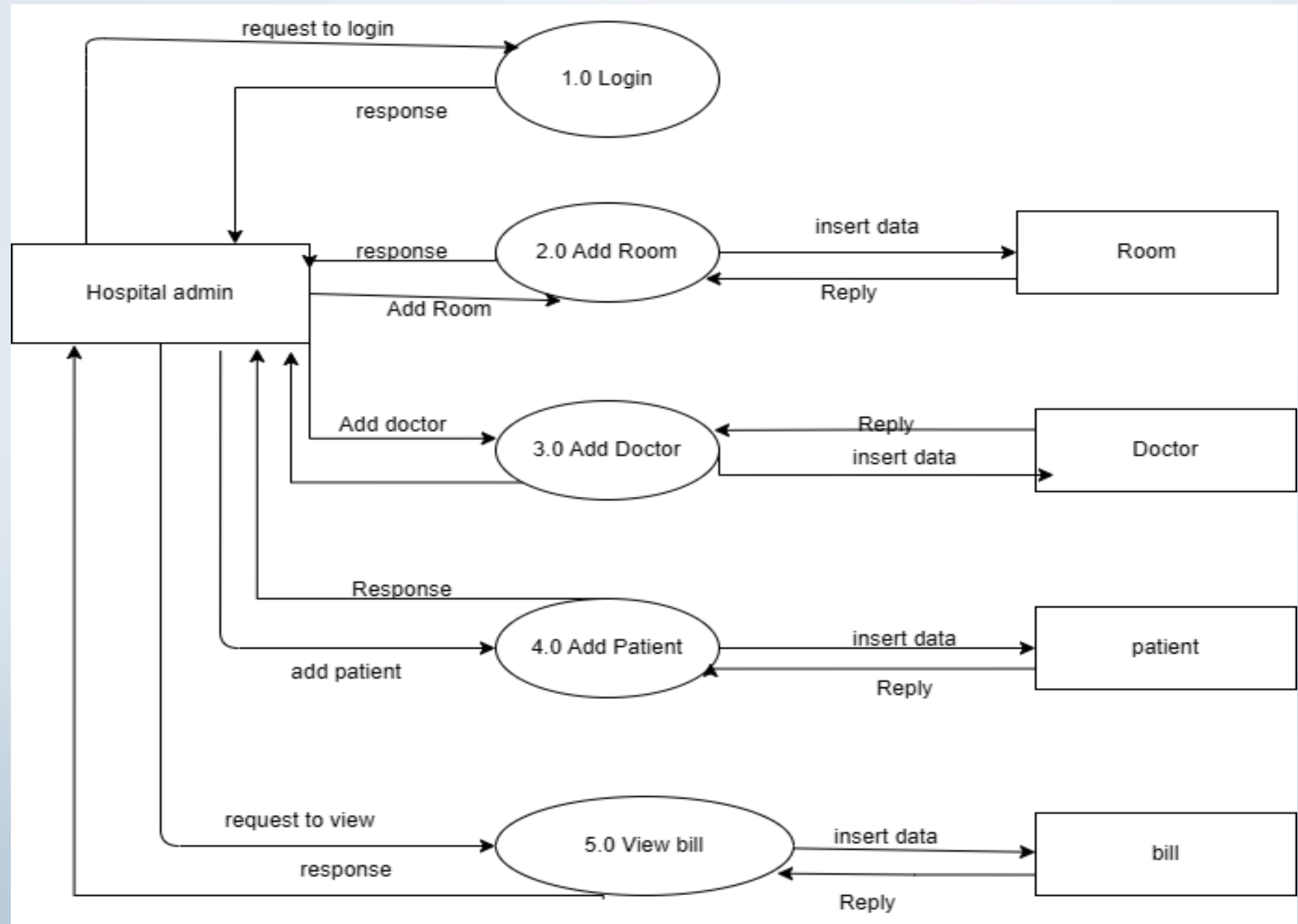


0th level DFD :



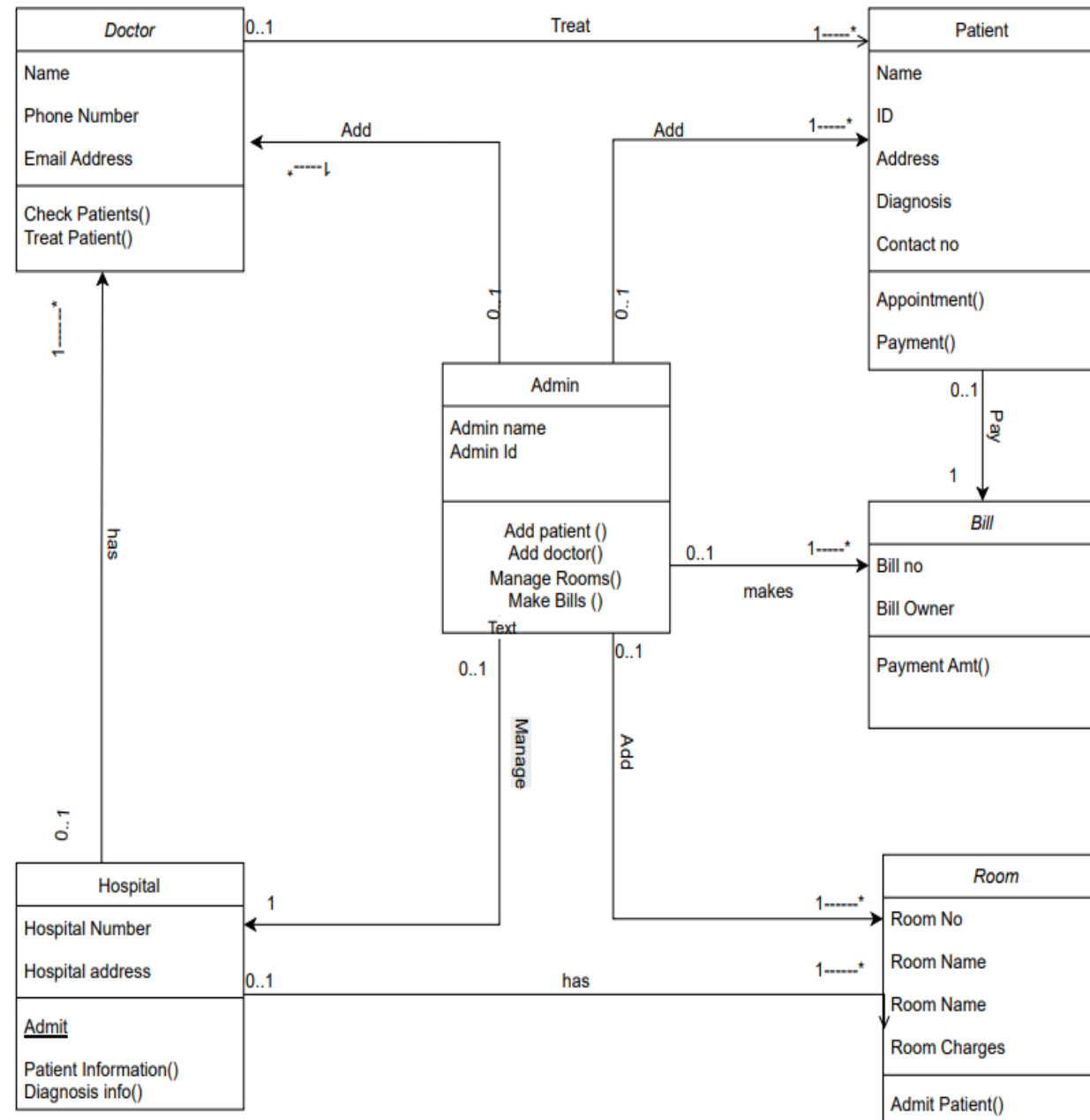


1st level DFD :



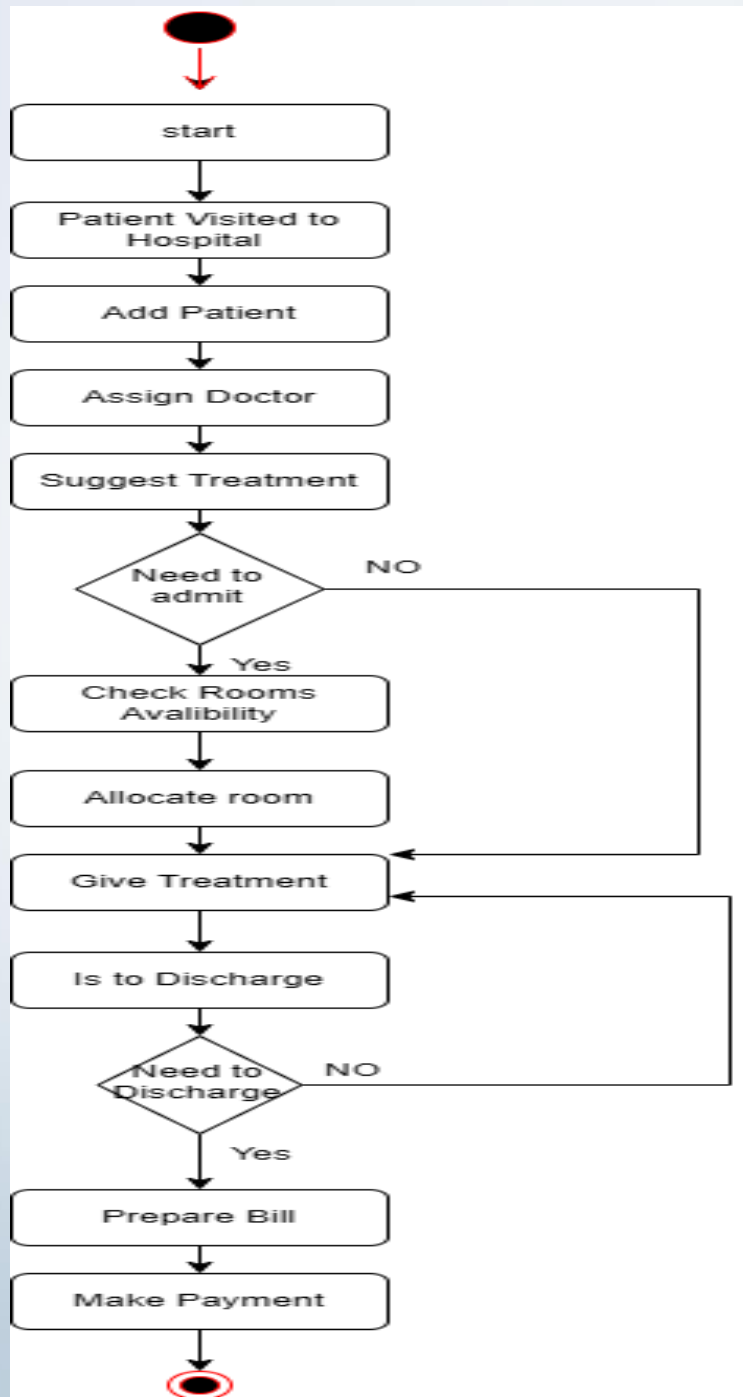


Class Diagram:



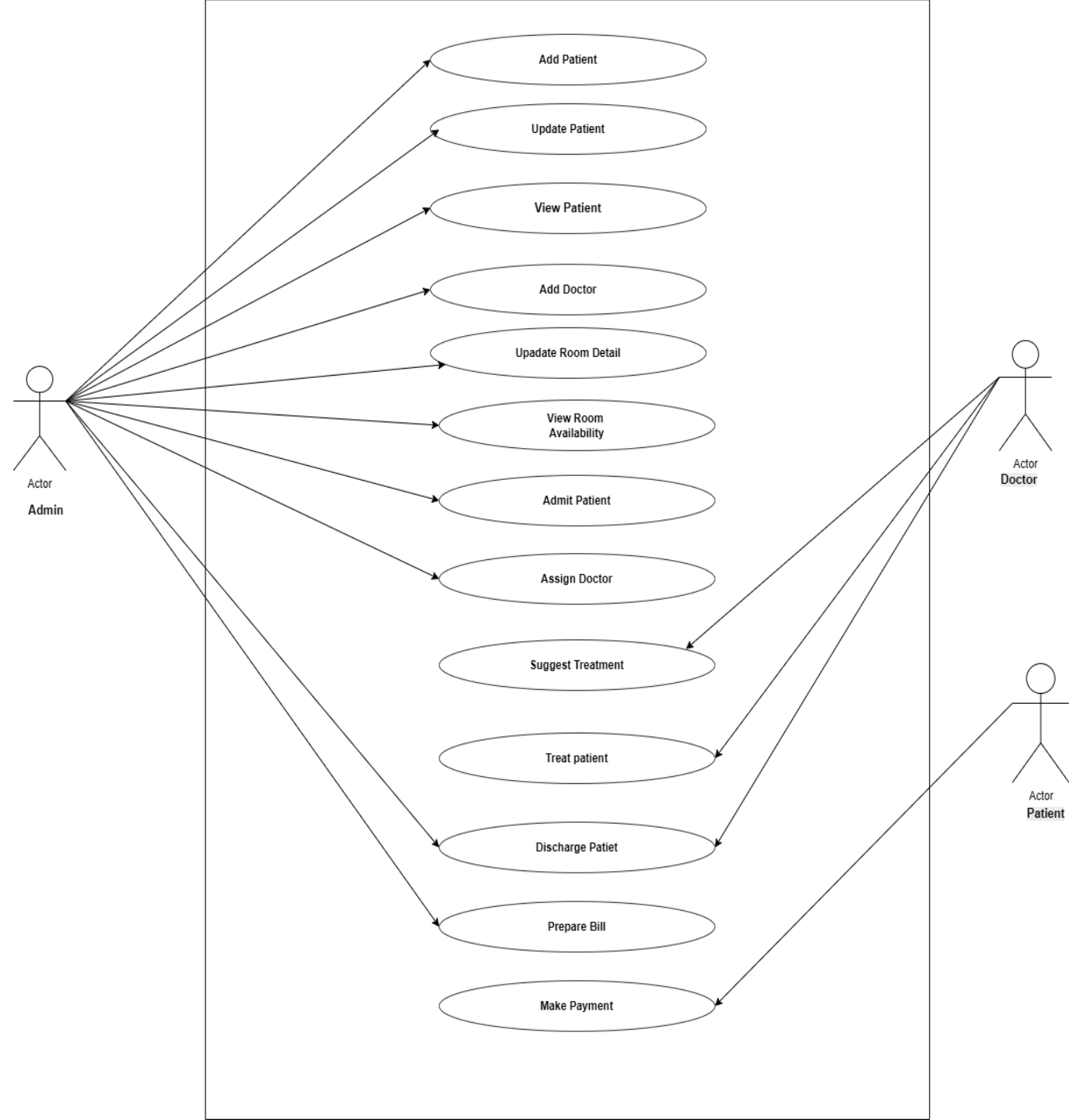


Activity Diagram:





Use Case Diagram:





Data Dictionary:

Add Patient Table:

Field name	Data type	constraints	size	Description
Patient ID	Int	Primery key	11	Patient ID
Patient name	Varchar	NULL	200	Patient name
Last name	Varchar	NULL	200	Last name
Contact no	Varchar	NULL	200	Contact no
Email-ID	Varchar	NULL	200	Email-ID
Age	Varchar	NULL	200	Age
Gender	Varchar	NULL	200	Gender
Blood Group	Varchar	NULL	200	Blood Group
Symptoms	Varchar	NULL	200	Symptoms
Diagnosis	Varchar	NULL	200	Diagnosis
Dr.ID	Int	NULL	200	Dr.ID
Dr.Name	Varchar	NULL	200	Dr.Name

Add Doctor Table:

Field name	Data type	constraints	size	Description
Doctor ID	Int	Primery key	11	Doctor ID
Dr.first name	Varchar	NULL	200	Patient name
Last name	Varchar	NULL	200	Last name
Contact no	Varchar	NULL	200	Contact no
Email-ID	Varchar	NULL	200	Email-ID
Age	Varchar	NULL	200	Age
Gender	Varchar	NULL	200	Gender
Qualification	Varchar	NULL	200	Qualification

Table: Admit Patient:

Field name	Data type	constraints	size	Description
Patient ID	Int	Primery key	11	Patient ID
Patient name	Varchar	NULL	200	Patient name
Gender	Varchar	NULL	200	Gender
Blood Group	Varchar	NULL	200	Blood Group
Disease	Varchar	NULL	200	Disease
Admit Date	Varchar	NULL	200	Admit Date
Room .	Varchar	NULL	200	Room .
Bed no	Varchar	NULL	200	Bed no
Dr.ID	Int	NULL	11	Dr.ID
Dr.Name	Varchar	NULL	200	Dr.Name

Table: Discharge Patient

Field name	Data type	constraints	size	Description
Patient ID	Int	Primery key	11	Patient ID
Patient name	Varchar	NULL	200	Patient name
Gender	Varchar	NULL	200	Gender
Blood Group	Varchar	NULL	200	Blood Group
Disease	Varchar	NULL	200	Disease
Admit Date	Date	NULL	8	Admit Date
Room .	Varchar	NULL	200	Room .
Bed no	Varchar	NULL	200	Bed no
Dr.ID	Int	NULL	11	Dr.ID
Dr.Name	Varchar	NULL	200	Dr.Name
Discharge Date	Date	NULL	8	Discharge Date

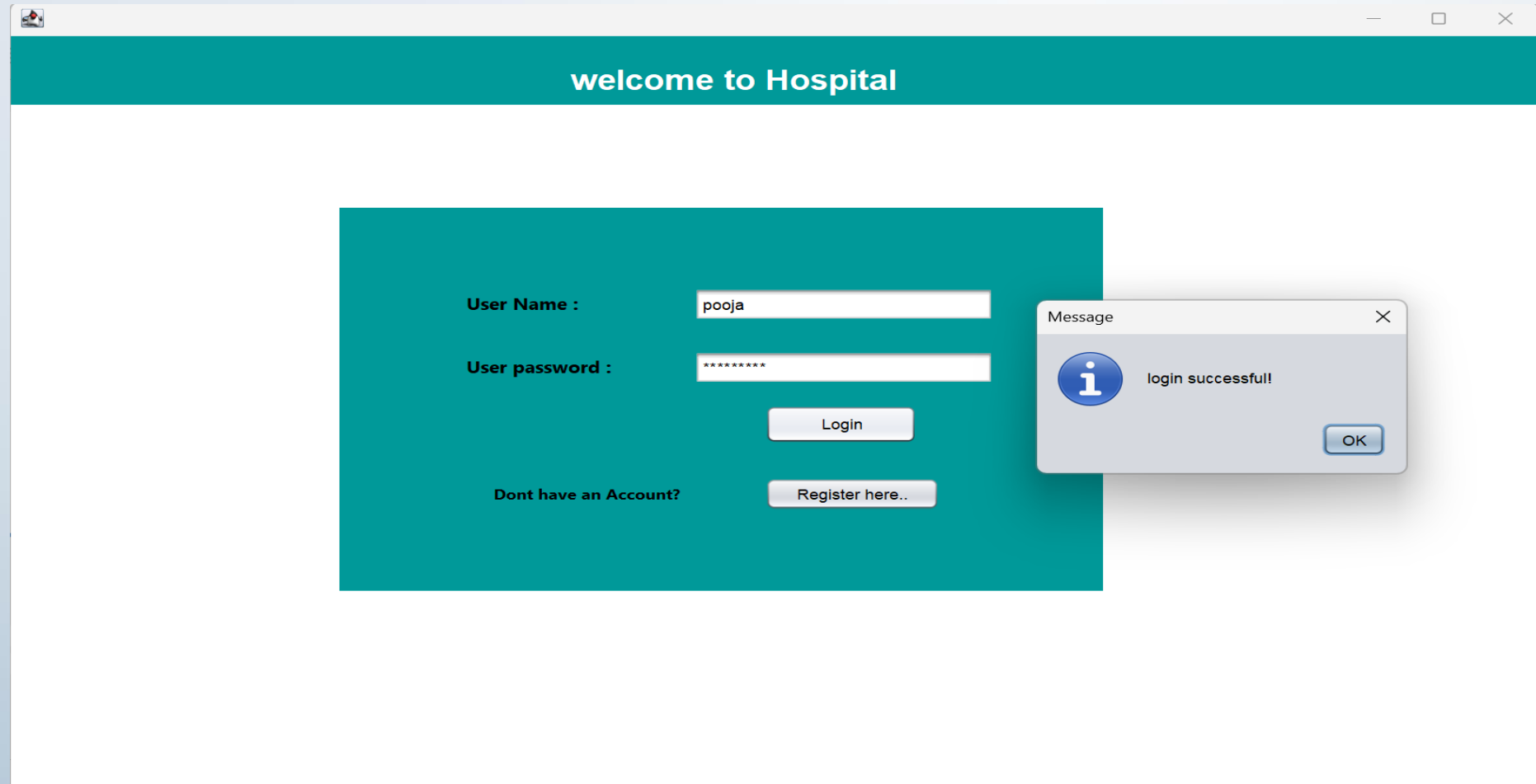
Table: Billing:

Field name	Data type	constraints	size	Description
Patient ID	Int	Primery key	11	Patient ID
Patient name	Varchar	NULL	200	Patient name
Gender	Varchar	NULL	200	Gender
Disease	Varchar	NULL	200	Disease
Admit Date	Date	NULL	8	Admit Date
Room .	Varchar	NULL	200	Room .
Discharge Date	Date	NULL	8	Discharge Date
Room charges	Varchar	NULL	200	Room charges
No.of days	Varchar	NULL	200	No.of days
Total Room charges	Varchar	NULL	200	Total Room charges
Treatment charges	Varchar	NULL	200	Treatment charges
Billing Date	Varchar	NULL	200	Billing Date
Payment Mode	Varchar	NULL	200	Payment Mode
Total Charges	Varchar	NULL	200	Total Charges
Bill No	Varchar	NULL	200	Bill No
Total Room charges	Varchar	NULL	200	Total Room charges



Input output screen

Login page :



The image shows a web browser window displaying a login page. The page has a teal header with the text "welcome to Hospital". Below the header is a teal login form. The form contains two input fields: "User Name :" with the value "pooja" and "User password :" with masked characters "*****". There are two buttons: "Login" and "Register here..". Below the "Login" button is the text "Dont have an Account?". A small dialog box titled "Message" is open on the right, showing an information icon and the text "login successful!" with an "OK" button.

welcome to Hospital

User Name : pooja

User password : *****

Login

Dont have an Account? Register here..

Message

login successful!

OK

REGISTRATION :

Registration

Admin Name :

pooja

Contact no :

9876543456

Address:

indapur

Email :

username

password

Message

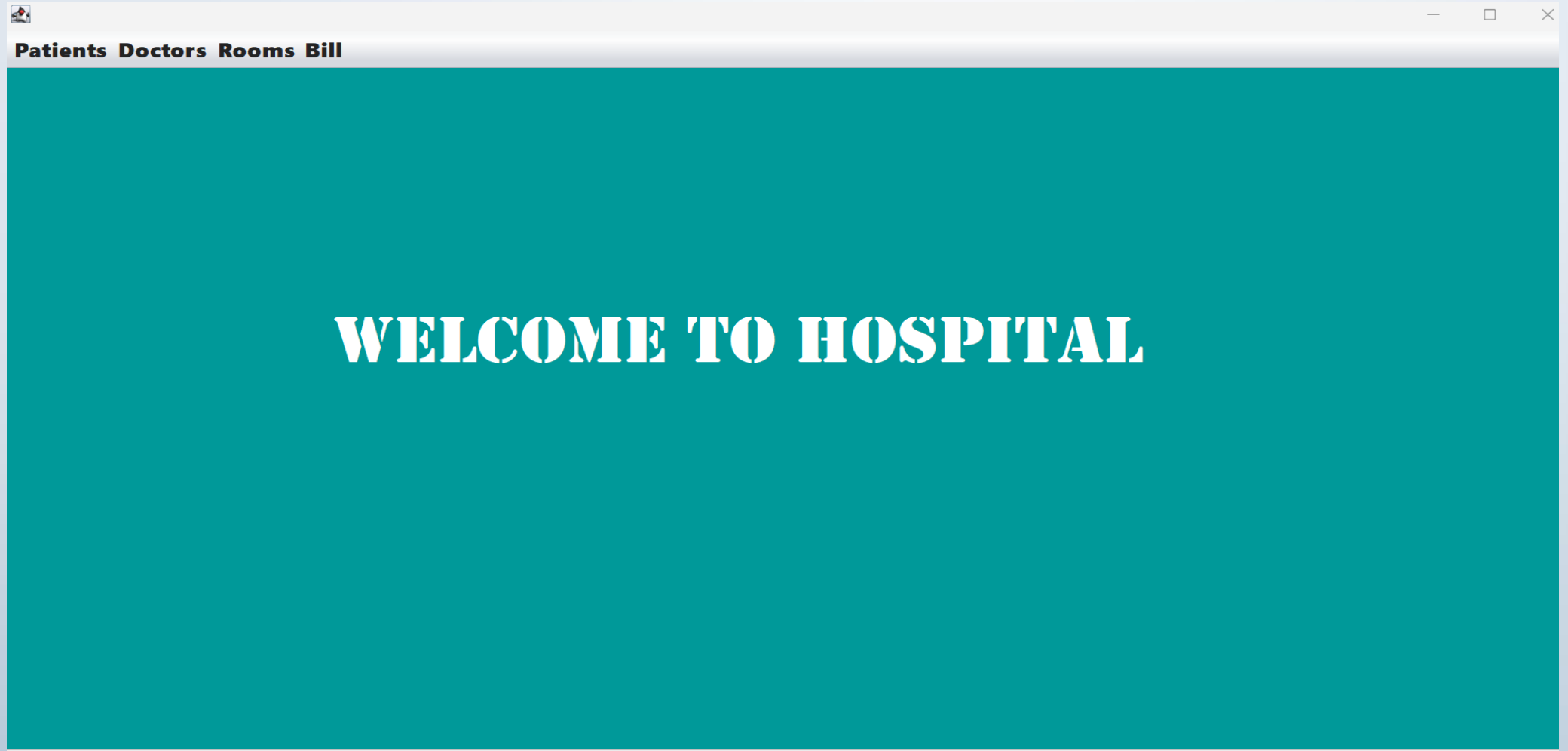
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registration successful!

OK

Register

Dashboard :



Add Patient Page

Patients Doctors Rooms Bill

Home

Patients Details

Patient ID

Patient name

Last name

Contact no

Email-ID

Age

Gender

Male

Blood Group

A+

Symptoms

Diagnosis

cold

Dr.ID

011

Dr.Name

Dr.Dhiraj Shi...

view

Add

Udate

delet

Clear

PatientID	PatientN...	Lastname	Contactno	EmailID	Age	Gender	Blood...	Symptoms	Diagnosis	Dr...	DrName
7	Adarsh	Raut	0987654...	ppoiuytrf	16	Male	B+	ear pain	ear infec...	11	Dr.Dhiraj Shingate
12	ajay cha...	chavan	0987654...	ajay123...	22	Male	O+	cold	cold	11	Dr.Dhiraj Shingate

Add New Doctor :

Patients Doctors Rooms Bill

Home

Doctor Details

Doctor Details

Doctor ID

Dr.first name

Dr.Last name

Contact no

Email-ID

Qualification

Age

Gender

Date of Joini...

view

Add

Udate

delet

Clear

DoctorID	firstname	lastname	Contactno	Email	Qualification	Age	Gender	DateJoining
11	Dhiraj	shingate	1234567890	dhiraj123...	bhms	34	Male	2025-10-29
12	Miss	shingate	0987654325	dhiraj123...	bhms	33	Male	2025-10-29

Message

i

All doctor records loaded successfully!

OK

Admit Patient :

— □ ×

Patients Doctors Rooms Bill

patient Admit Information

Patient Admit Info

Patient ID

Patient Name

Gender
Male ▾

Blood Group
A+ ▾

Disease
cold ▾

Admit Date

Room .
General... ▾

Bed no
Bed no1 ▾

Doctor ID
011 ▾

Doctor Name
Dr.Dhiraj Shi... ▾

view

Admit

Update

delet

Clear

PatientID	Name	Gender	Blood_Group	Disease	Admit_Date	Room	bedno	Doctor_ID
1	pooja	female	A+	headaches	2025-10-27	ICU	ICUBed no3	11
2	Aniket patil	Male	A+	fever	2025-10-27	ICU	ICU Bed no1	12
3	pratiksha	female	B+	stomachache	2025-10-27	ICU	ICU Bed no4	11
12	ajay chavan	Male	O+	cold	2025-10-27	ICU	ICU Bed no4	11
22	sakshi	female	AB+	malaria	2025-10-29	Genaral room	Bed no3	11

Message

×

All patient records loaded successfully!

OK

Discharge Patient :

Patients Doctors Rooms Bill

Discharge Information

Patient ID

search

Patient Name

Gender

Male

Blood Group

A+

Disease

cold

Admit Date

Room No.

Genaral...

bedno

Bed no1

Doctor ID

011

Doctor Name

Dr.Dhiraj Shi...

Discharge Date

view

Dis...

Update

delet

Clear

PatientID	Name	Blood_Gr...	Disease	AdmitDate	Room	bedno	DoctorID	DoctorNa...	Discharg...
1	pooja	A+	headaches	2025-10-27	ICU	ICUBed no3	11	Dr.Dhiraj ...	2025-10-31
12	ajay chavan	O+	cold	2025-10-27	ICU	ICU Bed n...	11	Dr.Dhiraj ...	2025-10-30
22	sakshi	AB+	malaria	2025-10-29	Genaral r...	Bed no3	11	Dr.Dhiraj ...	2025-10-30

Message

i

All patient records loaded successfully!

OK

Billing Page :

**Patients Doctors Rooms Bill**

Patients Details

NewSavePrint

Patients Details

Patient ID

12

search

Patient name

ajay chavan

Gender

Male

Diagnosis

cold

Admit date

27 Oct 2025

DD-MM-YYYY

Room

ICU

Discharge Date

1 Nov 2025

DD-MM-YYYY

Patient ID : 12

Patient Name : ajay chavan

Gender : Male

Diagnosis : cold

Room Type : ICU

Admit Date : 27 Oct 2025

Discharge Date : 1 Nov 2025

Room Charges : ₹ 5000.00

Treatment Charges : ₹ 1000

Total Amount : ₹ 6000.00

Payment Mode : online

Billing Date : 1 Nov 2025

Thank you for choosing Shingate Hospital!

We wish you a speedy recovery.

Room charges

5000.00

No.of days

5

Total Room charges

5000.00

Treatment charges

1000

Billing Date

1 Nov 2025

DD-MM-YYYY

PaymentMode

online

Total Charges

6000.00

Calculate

Advantages of System:-

- Immediate access of data.
- Friendly user interface.
- Time saving.
- Data can be easily insert update and delete.
- Saving paper work.
- Give facility of different types of enquiry.
- Data easily approachable.

THANK
YOU 🍊